

# AKHIL JAINI

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## SUMMARY

Computational Scientist and PhD Researcher with 6+ years of experience designing scalable real-time data-processing pipelines, high-performance software, and predictive statistical modelling for large-scale datasets. Bringing expertise in machine learning, Python and SQL analytics, and scientific computing, with a proven track record of extracting actionable insights from complex observational datasets.

## EDUCATION

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| <b>PhD in Computational Astrophysics</b><br>Swinburne University of Technology, Melbourne, Australia                                                         | <b>2023-Ongoing</b> |
| <b>Master of Technology in Astronomical Instrumentation</b><br>Indian Institute of Astrophysics, Bangalore, India and University of Calcutta, Kolkata, India | <b>2020-2022</b>    |
| <b>Bachelor of Engineering (Honours) in Electrical and Electronics</b><br>Birla Institute of Technology and Science – Pilani, Pilani, India                  | <b>2014-2018</b>    |

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++, CUDA C, SQL, Rust, HTML, JavaScript, TypeScript  
**Machine Learning:** PyTorch, Scikit-Learn, TensorFlow, NumPy, SciPy, Pandas  
**Software Engineering:** FastAPI, REST APIs, Docker, Git, GitHub Actions  
**Data & Infrastructure:** PostgreSQL, SQLite, Linux (Bash/Shell), AWS, HPC (OzStar/Slurm)

## RESEARCH IMPACT

First author of multiple peer-reviewed publications developing scalable computational methods and software adopted across international astronomy collaborations. **Full list of Publications:** [ui.adsabs.harvard.edu/public-libraries/6AWkqcbzTQWau\\_usyl5V8Q](https://ui.adsabs.harvard.edu/public-libraries/6AWkqcbzTQWau_usyl5V8Q)

## EXPERIENCE

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| <b>PhD Researcher – Centre for Astrophysics and Supercomputing, Swinburne University, Melbourne, VIC.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>June 2023-Present</b>       |
| <ul style="list-style-type: none"><li>Designed and developed predictive modelling and spatial cross-matching algorithms across multi-terabyte datasets using Python and SQL, improving feature localisation precision by 4x from ~1 to &lt;0.25 arcseconds and enabling reliable downstream analysis.</li><li>Built and deployed a real-time data processing pipeline for ingesting and processing streaming sensor data at &gt;20 TB/s throughput, developing anomaly detection algorithms to identify millisecond-scale extragalactic events.</li><li>Characterised and modelled complex time-series and polarimetric data using custom Python, SQL and statistical methods.</li><li>Collaborated within multidisciplinary international teams to translate complex analytics into production-grade astronomy software.</li></ul> |                                |
| <b>Research Software Engineer – Indian Institute of Astrophysics, Bangalore, India.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>August 2021-April 2023</b>  |
| <ul style="list-style-type: none"><li>Designed and validated data analysis workflows to identify and correct systematic errors in high-precision observational datasets, improving data quality and measurement reliability.</li><li>Developed Python and C++ control interfaces and data extraction pipelines to automate instrument calibration and control, high-precision spectrophotometric data acquisition, and end-user telemetry monitoring for a state-of-the-art space-based ultraviolet telescope.</li></ul>                                                                                                                                                                                                                                                                                                            |                                |
| <b>Computational Research Assistant – Raman Research Institute, Bangalore, India.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>January 2018-March 2020</b> |
| <ul style="list-style-type: none"><li>Designed and implemented a full-scale Python simulation framework to model multi-variable sensor networks for an orbiting radio interferometer, performing predictive risk analysis and scenario modelling under the rigorous operating conditions of outer space.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                |
| <b>Software Engineering Intern – National Centre for Radio Astrophysics, Pune, India.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>May 2017-July 2017</b>      |
| <ul style="list-style-type: none"><li>Developed a scalable GPU-accelerated high-throughput signal processing pipeline using CUDA C for real-time astronomical event detection and validated performance using simulated streaming data.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |

## PROJECTS

- **AstroConvert** (Python, Flet, AstroPy): Engineered a cross-platform application that intelligently detects, validates, and transforms heterogeneous scientific coordinate formats, reducing manual preprocessing through automated input recognition.
- **AI-Powered Bushfire Early Warning System** (PyTorch, FastAPI, Docker): Built an end-to-end predictive risk classification system analysing Sentinel-1 radar imagery to detect active fire fronts under cloud and smoke, deployed via FastAPI with an interactive dashboard.
- **Agentic Code Review & Document Analyst** (LangGraph, Ollama, SQLite): Developed a multi-agent AI system that autonomously analyses source code and technical documents, executes automated validation workflows, and generates structured review reports.